



Digital Forensics (SEng5307)

Chapter 1: Introduction to Digital Forensics



Introduction to Digital Forensics

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Introduction to Digital Forensics

- The world is becoming increasingly interconnected.
- The Internet is, however, a network of networks consisting of competing and concurrent technologies with users from different organizations and countries.
- Unfortunately for the investigator, the Internet was designed for robustness and redundancy, rather than security and traceability.
- Digital forensics is becoming increasingly important with the escalation of cybercrime and other network-related serious crimes.



Introduction to Digital Forensics

- Digital forensics is a critical field that applies scientific methods to gather, preserve, and analyze digital evidence.
- Understanding the laws and regulations governing electronic communications, cybercrimes, and data retention requires the continuous acquisition of new knowledge, methods, and tools.



Introduction to Digital Forensics

- Digital evidence plays a very important role in any criminal investigation.
- Therefore, CS, SE, and Computer security professionals need to acquire a fundamental understanding of digital forensics.
- This presentation aims to explore the key aspects of digital forensics, its processes, and its significance in crime investigations and legal proceedings.



Introduction to Digital Forensics

Forensic Science: is an application of scientific methods to establish factual answers to legal problems.

- It is an application of science in law.
- science that requires a creative mindset strictly adhering to established processes
- It is popular in fiction, contemporary media, and Crime shows.
- Interchangeably used with Criminalistics.



Introduction to Digital Forensics

Forensic Science: helps to establish facts related to questions such as:

- what has happened, how did it happen, who has been involved, and when did it occur?
- To solve such problems, a forensic scientist draws on methods and tools from a wide range of theoretical and applied sciences, including biology, medicine, physics, geology, computer science, and electrical engineering.





History of Forensic Science

Its history dates back to the 18th century.

- It contributed to a dramatic change in the effectiveness of law enforcement. Some of its contributions are:
 - Start of Forensic Toxicology.
 - Identification method through body measurements, and blood type characterization for crime characterizations.
 - Use of Fingerprints as a means of identification.
 - Designing principles for the application of science in investigations, document identification,
 - promoting the scientific method in criminal investigation



History of Forensic Science

Locard's Exchange Principle: states that “when a person or object comes in contact with another person or object, a cross-transfer of materials occurs”. Whenever two objects come into contact with one another, there is an exchange of materials between them.

- In this way, every criminal can be connected to a crime through trace evidence.
- It should, however, be noted that the principle cannot necessarily be directly applied to digital forensics as it is dynamic and different from physical evidence.





Crime Reconstruction

Referred as crime scene reconstruction is the process of determining the most likely hypothesis, or sequence of events, through the application of the scientific method. This can be done through a wide range of methods:

- Firearm ballistics tests,
- statistical simulations, and
- biological experiments.
- The objective is to establish a hypothesis about the event or sequence of events and then to test whether the hypothesis is possible or not.



Crime Investigations

Investigation is a systematic examination, typically to identify or verify facts. A key objective during investigations is to identify key facts related to a crime or incident and a common methodology used is 5WH (Stelfox, 2013):

- **Who:** Persons involved in the investigation, including suspects, witnesses, and victims.
- **Where:** The location of the crime and other relevant locations.
- **What:** Description of the facts of the crime in question
- **Why:** The motivation for the crime and why it happened at a given time.
- **How:** How the crime was committed.



Evidence Dynamics

is defined as “any influence that adds, changes, relocates, obscures, contaminates, or obliterates physical evidence, regardless of intent”.

It is any influence that adds, changes, relocates, obscures, contaminates, or obliterates evidence, regardless of intent.

- Helps to understand the actual behavior of evidence and plays an important role in crime scene reconstruction.
- Although the definition is originally intended for physical evidence, it is equally applicable to digital evidence.



Digital Forensics

is refers to forensic science applied to digital information, whereas a *digital investigation* refers to investigations in the digital domain.

- The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation, and presentation of digital evidence derived from digital sources to facilitate the reconstruction of events found to be criminal, or help to anticipate unauthorized actions shown to be disruptive to planned operations.



Digital Forensics

The role of digital forensics practitioners is to establish factual answers to legal problems.

Here are some of the processing of evidence and for the soundness of the analysis and its conclusions.

1. Crimes and Incidents: location of the incident is referred to as the *scene of the incident*, or in the case of a crime, the *digital crime scene*.

- can be initiated by real-time detection of cybersecurity incidents (e.g., intrusion detection) or as part of security incident handling processes.





Digital Forensics

2. **Digital Devices, media, and Objects:** are the center of digital forensics. They are physical objects that store digital data.

3. **Forensic Soundness and Fundamental Principles** refers to the systematic observation, measurement, and experiment, and the formulation, testing, and modification of hypotheses” It must satisfy:

- **Forensically Sound:** An investigation is forensically sound if it adheres to established digital forensics principles, standards, and processes.
- **Evidence Integrity:** Evidence integrity refers to the preservation of evidence in its original form.
- **Chain of Custody:** Chain of custody refers to the documentation of acquisition, control, analysis, and disposition of physical and electronic evidence.



Digital Forensics

4. **Crime Reconstruction in Digital Forensics:** can help test hypotheses about a possible chain of events. It includes:

- *Evidence examination:* Identify and characterize evidence relevant to an incident.
- *Role classification:* Examine the role of the evidence as a cause or effect of an event.
- *Event construction and testing:* Identify events and assess whether they are possible.
- *Event sequencing:* Combine events into event chains.
- *Hypothesis testing:* The hypothesis is tested using the scientific method.



Digital Evidence

Digital evidence is defined as any digital data that contains reliable information that can support or refute a hypothesis of an incident or crime.

- It is a good practice to process and store digital evidence in line with evidence integrity and chain of custody principles.
- manual process and detailed documentation of the chain of custody are nonetheless required.





Digital Evidence

Digital evidence can be in the context of:

- **Layers of Abstraction:** refers to the practice, used in all areas of computing, of hiding implementation details of higher layers of abstraction to reduce complexity.
- **Metadata:** is data about the data and is a valuable source of evidence in digital forensics as it contains information about data objects.
- **Error, Uncertainty, and Loss:** Errors, uncertainties, and data loss can affect the interpretation of timestamps, geographic location, and data authorship or ownership.



Challenges in Digital Forensics

- The complexity of digital systems and evolving technology.
- Issues with data integrity and evidence handling.



Recent Case Studies

- SpyEye case: An example of online bank fraud involving complex malware.
- Processes used in investigating multi-layered cybercrimes.



Legal Frameworks in Cybercrime

- Overview of international legal standards governing digital investigations.
- Importance of cooperation between law enforcement agencies.



The Future of Digital Forensics

- Emerging technologies and tools.
- Need for ongoing education and adaptation in forensic practices.





Conclusion

Digital forensics is a dynamic field crucial for investigating cybercrimes and managing digital evidence in legal contexts. A thorough understanding of its processes and challenges supports evidence integrity and strengthens legal proceedings.





Summary

I recommend exploring additional literature to gain a broader understanding of digital forensics.





Thank you very much!